

Power BI Dashboard Deployment & Usage Guide

This guide details how to build, model, and deploy the Olist Executive Dashboard using Power BI Desktop, utilizing the processed datasets and DAX formulas provided.

1. Importing Cleaned Data

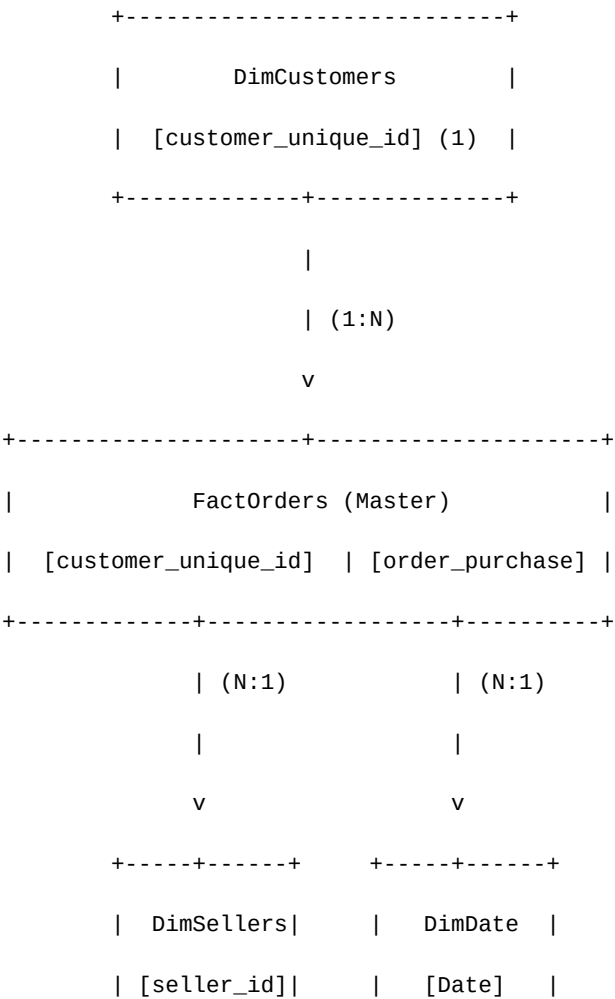
Import the following processed CSV files from the `data/processed/` directory into Power BI Desktop:

- 1. `master\_orders\_dataset.csv`
- 2. `olist\_customers\_dataset.csv`
- 3. `olist\_sellers\_dataset.csv`
- 4. `olist\_products\_dataset.csv`

Ensure all date fields are recognized as `Date/Time` types (e.g. `order\_purchase\_timestamp`, `order\_delivered\_customer\_date`).

2. Modeling Schema (Star Schema)

Navigate to the **Model View** and configure the relationships as shown below:



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Relationship Properties

- * **DimCustomers to FactOrders**:
- * Columns: `customer\_unique\_id`
- * Cardinality: One-to-many (1:*)
- * Cross filter direction: Single (DimCustomers filters FactOrders)
- * **DimSellers to FactOrders**:
- * Columns: `seller\_id`
- * Cardinality: One-to-many (1:*)
- * **DimDate to FactOrders**:
- * Columns: `Date` to `order\_purchase\_timestamp` (Date part)
- * Cardinality: One-to-many (1:*)

3. Creating DAX Tables & Measures

3.1 Create Calendar Table

Go to **Modeling > New Table** and enter the DAX formula for `DimDate`:

DimDate =

```
VAR MinDate = MIN(master_orders_dataset[order_purchase_timestamp])
```

```
VAR MaxDate = MAX(master_orders_dataset[order_purchase_timestamp])
```

```
RETURN
```

```
ADDCOLUMNS(
```

```
    CALENDAR(MinDate, MaxDate),
```

```
    "Year", YEAR([Date]),
```

```
    "Quarter", "Q" & FORMAT([Date], "Q"),
```

```
    "MonthNum", MONTH([Date]),
```

```
    "MonthName", FORMAT([Date], "MMMM"),
```

```
    "MonthYear", FORMAT([Date], "MMM YYYY"),
```

```
    "MonthYearSort", YEAR([Date]) * 100 + MONTH([Date]),
```

```
    "IsWeekend", IF(WEEKDAY([Date], 2) >= 6, "Weekend", "Weekday")
```

```
)
```

Note: Select the `MonthYear` column, and set "Sort by Column" to `MonthYearSort`.*

3.2 Implement Core Measures

Click **New Measure** on `FactOrders` and implement:

* **Total Revenue**: `Total Revenue = SUM(master\_orders\_dataset[revenue\_per\_order])`

* **Total Orders**: `Total Orders = DISTINCTCOUNT(master\_orders\_dataset[order\_id])`

* **AOV**: `Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)`

```

* **Late Delivery Rate**:  

Late Delivery Rate =  

DIVIDE(  

    CALCULATE([Total Orders], master_orders_dataset[late_delivery_flag] = 1),  

    CALCULATE([Total Orders], NOT(ISBLANK(master_orders_dataset[order_delivered_customer_date]))),  

    0  

)  

* **Repeat Purchase Rate (RPR)**:  

Repeat Purchase Rate =  

VAR MultiBuyers =  

    COUNTROWS(  

        FILTER(  

            VALUES(master_orders_dataset[customer_unique_id]),  

            CALCULATE(DISTINCTCOUNT(master_orders_dataset[order_id])) > 1  

        )  

    )  

RETURN  

    DIVIDE(MultiBuyers, DISTINCTCOUNT(master_orders_dataset[customer_unique_id]), 0)  

---

```

4. Page-by-Page Layout Deployment

Build out the 6 pages described in `dashboard/dashboard\_design\_guide.md` using standard visuals:

- * **Page 1 (Executive Summary)**: Focus on high-level KPI cards, the monthly revenue line chart, and the Class ABC revenue share donut chart.
- * **Page 2 (Sales)**: Set up the Map visual mapping regional revenues, and a clustered column chart comparing Category Revenue against Freight.
- * **Page 3 (Customer)**: Display customer growth trends and customer segment distributions.
- * **Page 4 (Seller Scorecard)**: Use a Scatter Plot mapping Seller Revenue vs. Seller rating, alongside a searchable matrix of low-performing sellers.
- * **Page 5 (Logistics)**: Plot lead time histograms and mapping LDR by customer state.
- * **Page 6 (Review)**: Map rating distributions and display the searchable comment table visual.